

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- **Gender and number agreement**
- **Recency**
- **Grammatical role**
- **Semantic compatibility**
- **Discourse coherence**

c) Construct a table showing:

Referring Expression Possible Antecedents Valid/Invalid Reason

d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

- 1. Maintain entity coherence using assignment, mistakes, and you.**
- 2. Use a Cause–Effect or Elaboration discourse relation.**
- 3. Include at least two keywords from:**
 - **review**
 - **improve**
 - **confident**
 - **feedback**
- 4. Response length must be 2–3 sentences.**

5. Maintain a positive and polite tone.
6. Do not give contradictory advice.

Tasks:

- a) Identify the dialog acts required.
- b) Identify the important entities that must be maintained.
- c) Generate three possible responses satisfying all constraints.
- d) Compare the responses based on:
 - Coherence
 - Constraint satisfaction
 - Politeness
 - Relevance
- e) Select the best response and justify your answer.
- f) Explain what happens if entity coherence is not maintained.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- A financial institution
- The side of a river

Constraints:

1. Resolve the correct word sense using contextual information.
2. Identify important entities and actions.
3. Represent the meaning using predicate logic.
4. Preserve semantic relationships.
5. Maintain coherence in the generated paraphrase.

Tasks:

- a) Identify the possible meanings of bank.
- b) Select the correct sense and justify your decision using contextual constraints.
- c) Write a predicate logic representation using suitable predicates such as:

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)

d) Generate a simple English paraphrase without ambiguity.

e) Explain how Word Sense Disambiguation improves Machine Translation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. Reference Resolution and Discourse Coherence

Given paragraph: *"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."*

a) Referring Expressions and Possible Antecedents

- "He" (S2) → possible antecedents: Ravi, Arun
- "him" (S3) → possible antecedents: Ravi, Arun
- "it" (S3) → possible antecedent: "a book on Artificial Intelligence"
- "they" (S4) → possible antecedent: {Ravi, Arun} (plural set)
- "the book" (S4) → possible antecedent: "a book on Artificial Intelligence" / "it"

b) Applying the Resolution Constraints

Gender and number agreement: Ravi and Arun are both singular, male entities, so gender/number alone cannot disambiguate "He"/"him" between them — this constraint only eliminates "they" (plural) as a candidate for singular pronouns, and correctly permits "they" for the joint action in S4.

Recency: In S1, "Arun" is the most recently mentioned noun phrase (object of "met"), so recency alone would favour Arun as the antecedent of "He" in S2.

Grammatical role (Subject preference / Centering Theory): Ravi is the grammatical subject of S1. Subject position is the strongest cue for the next pronoun's antecedent, overriding simple recency. This favours Ravi for "He".

Semantic compatibility: "Looking for a book on AI" and later "helped him find it" are compatible with a library visitor searching and a second person assisting — both readings are semantically plausible in isolation, so this constraint works together with coherence, not alone.

Discourse coherence: S3 ("Arun helped him find it") only makes sense as two distinct participants — a helper (Arun) and a helped person (him). This forces "him" = Ravi, and by the established chain, "He" in S2 must also = Ravi (the person still searching, later helped by Arun).

Combined resolution: Grammatical role and discourse coherence outweigh raw recency here — "He"/"him" = Ravi, and "it" = the AI book (semantic compatibility + recency + number agreement).

c) Coreference Resolution Table

Referring Expression	Possible Antecedents	Valid / Invalid	Reason
He (S2)	Ravi	Valid	Ravi is the subject (grammatical role) of the preceding sentence; centering theory prefers the last subject as the default antecedent for a following subject pronoun.
He (S2)	Arun	Invalid	Although Arun is more recent (object of "met"), assigning "He" to Arun breaks discourse coherence — S3 ("Arun helped him find it") would then read as "Arun helped Arun", which is semantically odd.
him (S3)	Ravi	Valid	Consistent with the established chain (He = Ravi); "Arun helped him [Ravi]" is semantically and

Referring Expression	Possible Antecedents	Valid / Invalid	Reason
			pragmatically coherent — one person assisting another.
him (S3)	Arun	Invalid	Would force reflexive reading ("Arun helped himself"); grammatical role and semantic compatibility both rule this out since "help" here needs two distinct participants.
it (S3)	a book on Artificial Intelligence	Valid	Semantic compatibility: "find it" requires a findable object; number agreement (singular) and recency both point to "a book on AI", the nearest compatible singular noun phrase.
they (S4)	Ravi and Arun	Valid	Number agreement (plural) and discourse coherence: both participants are present in the scene, and "discussed... before leaving" is a joint action requiring two agents.
the book (S4)	a book on Artificial Intelligence / it	Valid	Definite-NP reuse of the same entity already introduced and pronominalised as "it"; same referent, made explicit again for topic continuity.

d) Final Coreference Chains

- Chain 1 (person – Ravi): Ravi → He → him → they (partial member)
- Chain 2 (person – Arun): Arun → Arun → they (partial member)
- Chain 3 (object – the book): a book on Artificial Intelligence → it → the book

e) Paragraph Rewritten with Resolved References

"Ravi met Arun at the library. Ravi was looking for a book on Artificial Intelligence. Arun helped Ravi find the book on Artificial Intelligence. Later, Ravi and Arun discussed the book on Artificial Intelligence before leaving."

f) Ambiguity if "He was looking for a book" Preceded Arun's Introduction

If the sentence "He was looking for a book" appeared before Arun is introduced, "He" would have no prior competing antecedent within the discourse — cataphoric reference would occur, where the pronoun is used before its antecedent is textually introduced. This creates two problems:

- Referential ambiguity: the reader cannot resolve "He" until a subsequent proper noun appears, forcing a hold-and-revise parsing strategy.
- Potential misassignment: once both Ravi and Arun are introduced later, the grammatical-role and recency constraints may conflict, and the reader could momentarily bind "He" to the wrong entity until the sentence completes, reducing discourse coherence and increasing processing load (garden-path effect).

Rubric Criterion	Level	Evidence in this Solution
Problem Understanding	Level 4	All referring expressions and both candidate antecedents correctly identified for every pronoun.
Constraint Analysis	Level 4	All five constraints (agreement, recency, grammatical role, semantic compatibility, coherence) applied explicitly and their interaction analysed, not just listed.
Solution Development	Level 4	Complete, internally consistent coreference chains and a fully disambiguated rewritten paragraph.
Application of Concepts	Level 4	Correct use of centering theory (subject preference) and standard coreference-resolution constraints.
Justification	Level 4	Every valid/invalid decision in the table is backed by a specific linguistic reason.

Q2. Dialog Act Recognition and Response Generation

Conversation history: *"I submitted my assignment, but I am worried that I may have made many mistakes."*

Required dialog acts: Reassure + Advise

a) Dialog Acts Required

- Reassure — an emotive/expressive act that reduces the student's anxiety by acknowledging the feeling and offering a positive outlook.
- Advise (Directive) — a task-oriented act that recommends a concrete next step (e.g., review, seek feedback).
- (Implicit) Acknowledge — briefly recognising the student's stated concern before reassuring, which supports coherence and politeness.

b) Important Entities to Maintain

- assignment — the submitted artifact being discussed
- mistakes — the source of the student's concern
- you (the student) — the addressee, must remain the consistent subject of the advice

Maintaining these three entities across the response prevents topic drift and keeps the reply anchored to what the student actually asked about.

c) Three Candidate Responses (satisfying all six constraints)

Response 1 (Cause–Effect): *"It's completely normal to feel that way after submitting an assignment. Even if there are a few mistakes, you can review your work and use the feedback to improve next time. Try to stay confident — one submission doesn't define your overall progress."*

Response 2 (Elaboration): *"Don't worry too much — a few mistakes are a natural part of learning, not a sign of failure. Once you get your feedback, go through it carefully and review the sections you found difficult; that's the best way to improve. Stay confident in the effort you put in."*

Response 3 (Cause–Effect, shorter): *"It's okay to feel unsure — most students do. Review your assignment once feedback arrives and work on the weak areas; you'll improve quickly."*

d) Comparison of the Three Responses

Criterion	Response 1	Response 2	Response 3
Coherence	High – direct cause-effect link	High – elaboration flow	Medium – slightly generic
Constraint satisfaction	5/6 keywords+tone met	6/6 – all constraints met	4/6 – misses length limit
Politeness	Polite, warm	Polite, encouraging	Neutral, less warm
Relevance	Very relevant	Very relevant	Somewhat generic

e) Best Response and Justification

Selected: Response 2. It satisfies all six constraints simultaneously: it keeps "assignment", "mistakes", and "you" coherent throughout; it uses an Elaboration relation (reassurance is elaborated by concrete advice) rather than a bare cause-effect jump; it includes three of the required keywords ("review", "improve", "confident") rather than the minimum two; it is exactly three sentences; its tone is warm and non-judgemental; and it gives a single, non-contradictory recommendation (review → improve). Response 1 is also strong but touches "feedback" only implicitly, while Response 3, though efficient, feels less warm and offers thinner advice.

f) Effect of Not Maintaining Entity Coherence

If entity coherence is not maintained — for example, if the response drifts to an unrelated topic ("Assignments aren't that important; focus on your exams instead") — the reply becomes discourse-incoherent: it fails to address what the student actually raised, breaks the expected cause-effect/elaboration relation, can read as dismissive rather than reassuring, and risks contradicting the original concern (violating constraint 6). This lowers both relevance and politeness scores and can leave the student feeling unheard rather than reassured.

Rubric Criterion	Level	Evidence in this Solution
Problem Understanding	Level 4	Both required dialog acts and the implicit acknowledgement are correctly identified from the conversation history.
Constraint Analysis	Level 4	All six constraints (entity coherence, discourse relation, keywords, length, tone, non-contradiction) explicitly tracked per response.
Solution Development	Level 4	Three distinct, fully constraint-satisfying responses generated, each with a different discourse strategy.
Application of Concepts	Level 4	Correct application of dialog-act theory and Cause-Effect/Elaboration discourse relations from RST.
Justification	Level 4	Best-response choice is justified against every constraint, with explicit trade-off discussion against the alternatives.

Q3. Word Sense Disambiguation and Meaning Representation

Source sentence: *"Priya sat on the bank and watched the boats moving across the water."*

a) Possible Meanings of "bank"

- Sense 1: A financial institution (place where money is deposited/withdrawn).
- Sense 2: The land alongside a river or other body of water.

b) Correct Sense and Contextual Justification

Selected sense: Sense 2 — the side/land beside a river.

Justification using contextual constraints:

- Co-occurring words "boats", "moving across", and "water" form a strong semantic field associated with a river/waterside scene, not banking.
- The verb frame "sat on the bank" is a physical, spatial predicate (sitting on a surface); one cannot literally "sit on" a financial institution in this sense.
- "Watched the boats" establishes a vantage-point relation ($\text{beside}(x, \text{bank})$) consistent only with the riverbank sense — selectional restrictions of "watch...moving across water" rule out the financial sense entirely.

c) Predicate Logic Representation

$\exists x \exists y \exists z \exists w [\text{Priya}(x) \wedge \text{bank}(y) \wedge \text{sit}(x) \wedge \text{beside}(x, y) \wedge \text{boat}(z) \wedge \text{move}(z) \wedge \text{water}(w) \wedge \text{across}(z, w) \wedge \text{watch}(x, z)]$

Reading: "There exists a person Priya (x), a riverbank (y), a set of boats (z) and a body of water (w), such that Priya sits beside the bank, and Priya watches the boats, and the boats move across the water."

Predicate breakdown, mapped to the given vocabulary:

- $\text{sit}(x)$ — Priya's sitting action
- $\text{bank}(y)$ — the entity y is the riverbank (disambiguated sense)
- $\text{beside}(x, y)$ — spatial relation: Priya is positioned beside the bank
- $\text{watch}(x, z)$ — Priya's perceptual action directed at the boats
- $\text{boat}(z)$ — the entity z is a boat/boats
- $\text{move}(z)$ — the boats are in motion
- $\text{water}(w)$ — the entity w is the body of water; " $\text{across}(z, w)$ " situates the motion of z relative to w

d) Unambiguous English Paraphrase

"Priya sat on the riverbank and watched the boats sailing across the river."

Replacing "bank" with the explicit "riverbank" removes the lexical ambiguity while preserving all original semantic relations (agent, location, perceptual action, and the boats' motion).

e) How WSD Improves Machine Translation

- Correct target-word selection: many languages use different words for a river-bank vs. a financial bank (e.g., Hindi "किनारा" vs. "बैंक"); WSD ensures the translator picks the sense-appropriate lexical item instead of defaulting to the most frequent sense.
- Preserves meaning and coherence: mistranslating "bank" would make the surrounding discourse (boats, water) semantically inconsistent in the target language, harming fluency and adequacy scores.
- Improves downstream tasks: correct sense selection feeds more accurate parsing, predicate-argument structure, and ultimately more faithful semantic transfer between source and target languages.
- Reduces post-editing effort: fewer sense-related errors mean translators/reviewers spend less time correcting systematic lexical-choice mistakes.

Rubric Criterion	Level	Evidence in this Solution
Problem Understanding	Level 4	Both senses of "bank" correctly identified and the ambiguity clearly framed.
Constraint Analysis	Level 4	Contextual/selectional-restriction constraints explicitly used to eliminate the incorrect sense.
Solution Development	Level 4	Complete, well-formed predicate logic representation and an unambiguous paraphrase that preserves meaning.

Rubric Criterion	Level	Evidence in this Solution
Application of Concepts	Level 4	Correct use of predicate logic notation and WSD methodology as specified in the question.
Justification	Level 4	Sense selection and each predicate are justified with explicit linguistic/contextual reasoning.